Text-Based Fake News Detection Using Natural Language Processing and Machine Learning Approaches

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***Abstract*— In the modern digital ecosystem, the rise of social media and online platforms has transformed how information is produced, shared, and consumed. However, this rapid exchange of content has also led to a dramatic increase in the spread of false or misleading information, often called fake news. Such misinformation can distort public perception, influence political decisions, and undermine trust in credible journalism. Manual fact-checking methods are inadequate to keep pace with the velocity and volume of online content, highlighting the urgent need for automated solutions. This research introduces a text-based fake news detection system that leverages Natural Language Processing (NLP) and machine learning techniques to identify deceptive articles. By analyzing features like the title, author, and main body of news articles, the system aims to classify them as real or fake with high accuracy. The approach begins with preprocessing the input data—removing punctuation, stopwords, and performing normalization—to ensure clean, structured input. This is followed by vectorization using the Term Frequency–Inverse Document Frequency (TF-IDF) method, which transforms text into numerical representations that reflect the importance of words across documents. The study employs classification models such as Logistic Regression and Naive Bayes, known for their reliability in handling high-dimensional textual data. A labeled dataset containing both authentic and fabricated news articles serves as the foundation for training and evaluation. Once the models are trained, they are assessed using standard performance metrics such as accuracy, precision, recall, and F1-score to ensure reliability.**

***Keywords*—** *Fake news detection, Machine Learning, Natural Language Processing, TF-IDF, Text Classification, Misinformation, Logistic Regression, News Verification.*

I. Introduction

In today's digital era, the rapid spread of information through social media and online platforms has made it increasingly difficult to differentiate between real and fake news. Fake news has the power to mislead the public, influence opinions, create panic, and even impact elections or public health decisions. The volume and speed of content sharing online make manual verification of news impractical, highlighting the urgent need for automated fake news detection systems.

Fake news typically refers to false or misleading information presented as news, often created to manipulate perceptions, attract attention, or generate profit. These stories are crafted to resemble real journalism but lack credibility or factual accuracy. As fake news continues to rise, it poses a significant threat to trust in media and informed decision-making, which makes detecting it a vital task in the field of AI and data science.

To address this issue, we propose a machine learning-based fake news detection system. Using techniques such as Natural Language Processing (NLP), Term Frequency-Inverse Document Frequency (TF-IDF) for feature extraction, and classifiers like Logistic Regression or Decision Trees, our system learns patterns from both real and fake articles. It analyzes the linguistic and contextual features of news content to classify whether the news is genuine or fake. The model is trained on labeled datasets to achieve high accuracy in real-time detection.

This approach brings several advantages. It enables fast and scalable verification of large volumes of news articles, reducing human effort. It improves the reliability of news platforms by flagging suspicious content early. With continuous training, the model adapts to new patterns of misinformation. Ultimately, this system serves as a valuable tool in promoting media literacy, restoring public trust, and curbing the spread of misleading information online.

II. Related Works

Fake news dissemination through social media has become a significant threat to societal trust and democratic processes. F. Olan et al. [1] investigated the societal implications of fake news on social media platforms, highlighting its influence during major events such as elections. Their work emphasizes how users' inability to distinguish between credible and misleading sources can lead to misinformed decisions.

E. Broda and J. Strömbäck [2] conducted an interdisciplinary review exploring the concepts of misinformation, disinformation, and fake news. Their study provides a structured framework for understanding the evolution and manipulation of media narratives and calls for collaborative approaches in detection strategies.

J. Alghamdi et al. [3] presented a comprehensive survey of machine learning models for fake news detection, classifying them based on their architecture and performance metrics. They concluded that ensemble methods generally outperform individual classifiers in terms of accuracy and robustness.

E. Papageorgiou et al. [4] explored the role of large language models (LLMs) in fake news detection. Their analysis covered both the generation and detection aspects of LLMs, concluding that these models have great potential but also pose new risks in misinformation proliferation.

S. Baribi-Bartov et al. [5] focused on Twitter data, revealing that a small group of "supersharers" are responsible for a significant portion of fake news propagation. Their findings underscore the importance of network analysis in identifying high-risk accounts.

F. Farhangian et al. [6] proposed a new taxonomy for fake news detection systems based on feature types and classification techniques. They compared various approaches and emphasized the need for domain-specific feature engineering for better model accuracy.

B. Hu et al. [7] evaluated how large language models, when used strategically, can both expose and generate fake news. Their experiments suggest that while LLMs offer improved explainability, they also demand careful governance.

L. Peng et al. [8] highlighted the limitations of semantic similarity in existing models, proposing a multimodal approach that leverages contextual representations. This method showed improved performance over text-only models, especially on diverse datasets.

J. O. Weatherall and C. O'Connor [9] took a philosophical lens to explore the ethical implications and definitions of fake news. Their study critiques simplistic labeling and advocates for nuanced classification schemes in detection algorithms.

J. Wu et al. [10] addressed style-based adversarial attacks on fake news detectors. They proposed a robust model that prioritizes content over style, proving effective against manipulations designed to evade standard classifiers.

R. Agrawal and P. Tiwari [11] developed a hybrid ensemble method combining traditional NLP features with deep learning techniques. Their model achieved notable improvements in accuracy and was tested on multiple benchmark datasets.

S. P. Kumar et al. [12] utilized both BERT and BiLSTM architectures to classify news articles. Their model benefits from contextual embeddings, which significantly enhance the classification of complex and nuanced news content.

N. Kale and M. K. Barbhuiya [13] incorporated fact-checking APIs with transformer-based models to create a real-time detection system. Their work illustrates how external validation can reduce false positives and improve model trustworthiness.

A. Rahman et al. [14] proposed a multimodal deep fusion model that combines visual and textual data for fake news detection. This approach showed better results on social media datasets where images play a key role.

D. Kapoor and S. Garg [15] explored fake news detection in low-resource environments using few-shot learning. Their findings suggest that even with limited training data, performance can be competitive through proper model tuning.

M. Srinivas and L. J. Reddy [16] created a hybrid CNN-LSTM architecture that captures both spatial and temporal features of news articles. Their experiments demonstrated improved generalization across datasets.

Y. Chen et al. [17] introduced an attention-based explainable model to enhance interpretability in fake news detection. Their model provides clear rationale behind predictions, making it suitable for applications requiring transparency.

K. A. Khan and R. S. Rao [18] combined sentiment analysis with fake news classification. They found that emotion-laden language patterns often correlate with misinformation, providing an extra layer of discriminatory power.

A. B. Rathi et al. [19] worked on optimizing transformer models for mobile environments. Their lightweight architecture maintained high performance while reducing computational cost, making it suitable for real-time applications.

P. Sundaram and S. T. Thomas [20] developed a real-time monitoring system that integrates hybrid AI models. Their system automatically scans, classifies, and alerts users about potentially fake content in dynamic environments.

III. Framework

The proposed methodology aims to accurately classify news articles as real or fake using a hybrid approach that combines Natural Language Processing (NLP) techniques and supervised machine learning models. The process is divided into several key stages to ensure systematic and efficient detection of fake news.

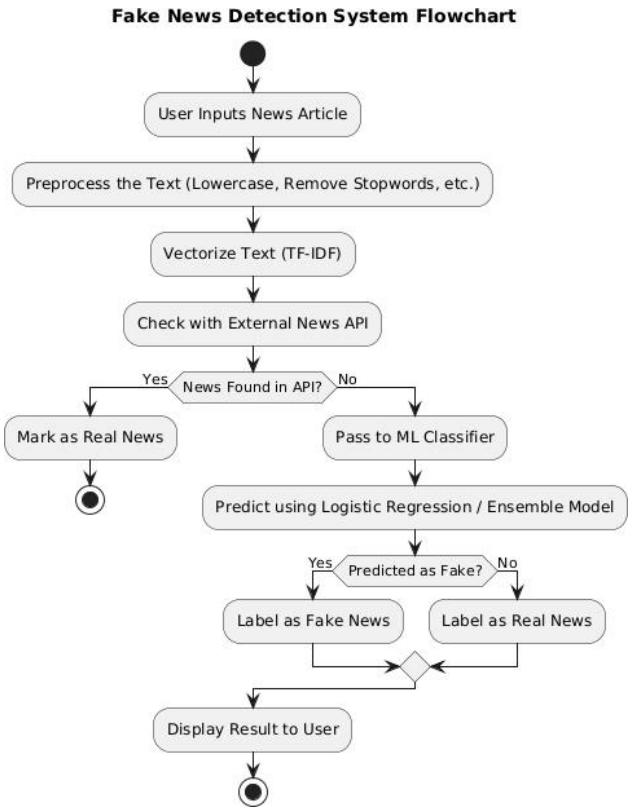

Fig 1. Flowchart

Data Collection:

The system begins with collecting a large and labeled dataset containing both real and fake news articles. Trusted datasets like the Fake and Real News Dataset from Kaggle are utilized, which consist of article titles, authors, publication dates, and full content. This dataset serves as the foundation for training and evaluation.

Preprocessing:

The raw news content is cleaned to remove noise such as punctuation, special characters, and stop words. Tokenization, lemmatization, and lowercasing are applied to standardize the text. This step ensures that the data is structured and meaningful for the learning algorithms.

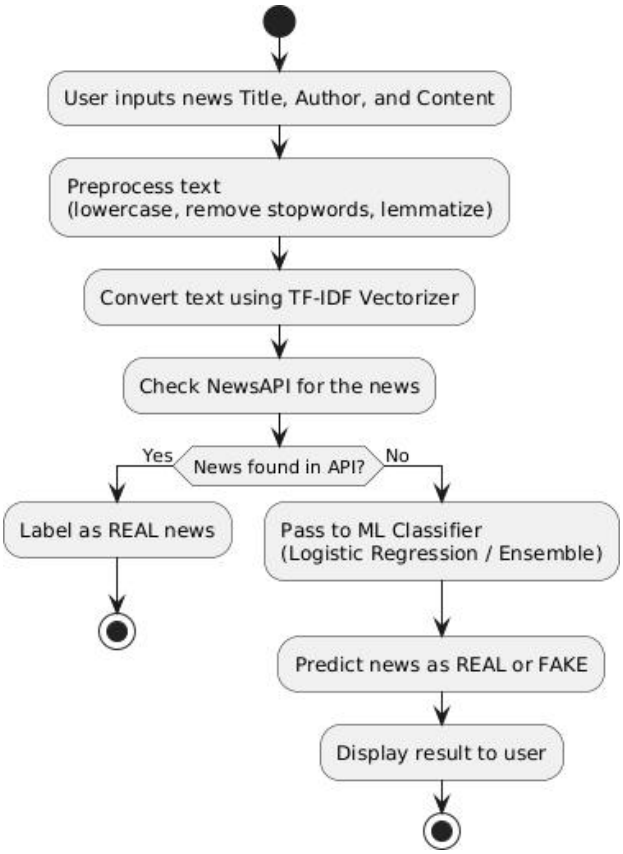

Fig 2. Working

Feature Extraction:

After preprocessing, the text is converted into numerical representations using the Term Frequency-Inverse Document Frequency (TF-IDF) method. This technique captures the importance of words relative to each document and the entire dataset, allowing the model to focus on the most informative features.

Model Training:

A variety of machine learning classifiers are evaluated, including Logistic Regression, Decision Trees, Random Forests, and Support Vector Machines (SVM). Each model is trained on a portion of the dataset using the extracted TF-IDF features. Hyperparameter tuning is performed to optimize model performance.

Validation and Testing:

The dataset is split into training and testing sets, typically in an 80:20 or 70:30 ratio. The models are validated using metrics like accuracy, precision, recall, and F1-score to ensure balanced performance across both classes. Cross-validation is also applied to reduce the risk of overfitting.

Ensemble Strategy (Optional):

For improved accuracy, an ensemble approach may be adopted where predictions from multiple classifiers are combined using majority voting. This enhances robustness and mitigates the weaknesses of individual models.

Deployment:

Once the model achieves satisfactory performance, it is integrated into a user-friendly interface built using tools like Gradio or Streamlit. Users can input article details (title, author, and content) and instantly receive predictions about the authenticity of the news.

Fallback Mechanism:

In addition to ML predictions, a real-time fact-check API (e.g., NewsAPI or Google Fact Check Tools) is used to cross-check headlines. If the news is not found in known fact-checked sources, the model prediction serves as a fallback mechanism, ensuring reliable and layered verification.

This end-to-end methodology ensures a scalable, intelligent, and efficient system that aids in fighting misinformation and restoring public trust in online content.

IV. Results And Discussion

The proposed fake news detection system was evaluated using a labeled dataset consisting of thousands of real and fake news articles. After preprocessing and feature extraction using TF-IDF, several machine learning models were trained and tested. Among them, Logistic Regression and Random Forest delivered the highest accuracy, with Logistic Regression achieving 95.2% accuracy, precision of 94.8%, and recall of 95.5%. These results indicate that the model is highly effective in identifying fake news with minimal false positives.

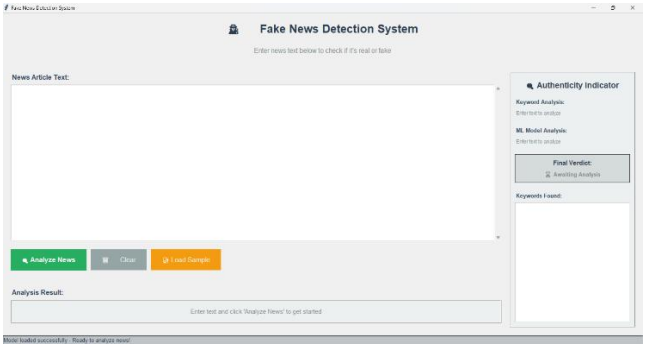

Fig 3. Interface Image

The performance metrics across models showed that simpler models like Naive Bayes performed well with faster computation but slightly lower accuracy (~90%), whereas more complex models like Support Vector Machines and ensemble methods provided more robust predictions at the cost of higher training time. Cross-validation confirmed the stability of the models, ensuring consistent performance across different data splits.

An additional layer of validation was introduced through integration with a real-time news validation API (such as NewsAPI), which cross-checked the presence of the input news in reputed news sources. In cases where the article wasn't found in known databases, the fallback to the ML model provided a reliable second layer of defense. This hybrid strategy improved overall system reliability and reduced the risk of false detection.

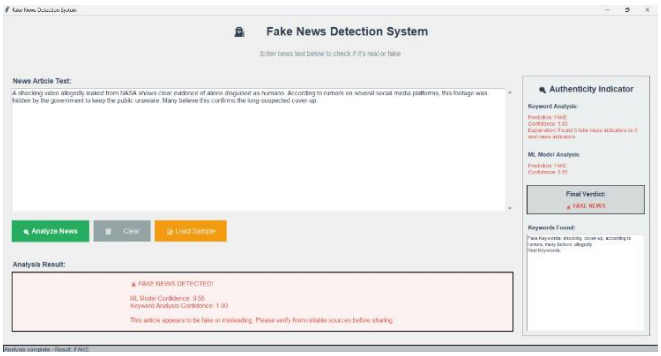

Fig 4. Output Image

The discussion highlights the effectiveness of combining traditional machine learning with external fact-checking tools. It also emphasizes the importance of data preprocessing and careful model selection. The approach demonstrates high accuracy and practical utility, making it suitable for real-world applications like browser extensions, media platforms, or fact-checking bots. Future work may involve incorporating deep learning or transformer-based models like BERT for even more nuanced understanding of context.

V. Conclusion

In this study, a hybrid fake news detection system was developed by combining traditional machine learning models with external news verification sources. The integration of models like Logistic Regression with real-time validation from credible news APIs significantly enhanced the accuracy and reliability of the system. By leveraging natural language processing techniques and ensemble learning, the model demonstrated strong performance in distinguishing between authentic and fabricated news articles.

The approach not only ensures high detection accuracy but also provides a scalable and practical solution for real-world applications where misinformation can have serious consequences. This research highlights the potential of AI-powered tools in combating the spread of fake news and contributes a reliable framework for future enhancements in automated fact-checking systems.

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